

UNIT-2

Linux Commands (File Manipulation, Redirection, Streams, grep, cut, awk, Command Chaining)

Scenario 1 — Student Records Folder Setup

Your department maintains student academic records in the Linux system.

You are placed in the *IT-Lab-03* folder and asked to organize files.

Tasks

1. Check what files exist in your current folder.
 2. Move into the folder named Sem-6.
 3. Create a new empty file named records.txt.
 4. View the content of records.txt (verify it is empty).
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Scenario 2 — Appending Results from Terminal

A faculty member gives the following student details to be appended into records.txt:

RegNo: 1023 | Name: Arjun | Grade: A

Tasks

1. Append the above line to records.txt using a redirection operator.
 2. Now overwrite the entire records.txt so that it contains only:
RegNo: 1024 | Name: Ravi | Grade: B
 3. Create a file named filelist.txt that stores the output of listing all the files in the current directory.
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Scenario 3 — Debugging Execution Logs (STDIN, STDOUT, STDERR)

During compilation of a C program, the system produces three different types of messages.

Tasks

1. Describe **STDIN**, **STDOUT**, **STDERR** in your own words.
2. Give one example command for each:
 - For reading input from user (STDIN)
 - For printing normal output (STDOUT)
 - For displaying an error (STDERR)

(Example expected: Students may use cat, echo, invalid commands, etc.)

Scenario 4 — Searching Inside Text with grep

The Dean asks you to check whether student records contain the keyword "**Grade**".

Tasks

1. Search in records.txt for lines containing "Grade".
2. Display only those lines that **do not** contain "A".
3. Apply a pipe to:
 - Search lines containing "Grade"
 - Then filter again for "B"

Expected form:

grep "Grade" records.txt | grep "B"

Scenario 5 — Extracting Specific Fields using cut

Your file contains data in the format:

RegNo: 1024 | Name: Ravi | Grade: B

Tasks

1. Extract only the **student name** (2nd word after spaces) using cut.
2. Extract the **first 5 characters** from every line in the file.

Scenario 6 — Working with Fields using awk

Your system stores multiple student details inside records.txt.

Tasks

1. Print only the **third field** (Grade value) from the file.
2. Count how many total lines are present in the file using awk (Hint: NR).

Scenario 7 — Data Pipeline Automation (Command Chaining)

Suppose records.txt contains multiple entries like:

RegNo: 1024 | Name: Ravi | Grade: B

RegNo: 1023 | Name: Arjun | Grade: A

RegNo: 1028 | Name: Kiran | Grade: A

The principal wants to extract **only the first three characters of the student names of those who secured Grade A.**

Task

Construct a **single Linux pipeline command** that will:

1. Filter lines with "A"
2. Extract the field containing the name
3. Print only the **first 3 characters** of names

(Expected chain uses grep | cut | awk or variations)
